

Kubernetes Secret Not Accessible SOP

Objective

Restore application access to Kubernetes secrets when pods cannot read required credentials, tokens, or certificates.

Scope

Applicable to Kubernetes workloads, platform operations, DevOps teams, SRE teams, application teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

- 1. Access to Kubernetes cluster using kubectl and the correct namespace context.
- 2. Access to application, pod, deployment, service, ingress, and node logs where applicable.
- 3. Access to monitoring dashboards such as Grafana, Prometheus, cloud monitoring, or container insights.
- 4. Permission to inspect manifests, secrets, configmaps, RBAC bindings, PVCs, and rollout history.
- 5. Escalation contact for DevOps, platform, infrastructure, network, and application teams.

Incident Metadata

Field	Value
Ticket ID	T029
Issue	Kubernetes secret not accessible
Category	Security
Service	Kubernetes
Severity	Medium
Status	Resolved
Description	Application cannot read secret
Expected Resolution	Check RBAC permissions and mount path

Procedure

- 1. Confirm secret name, namespace, key name, consuming pod, and failure message.
- 2. Verify the secret exists and contains the expected keys.
- 3. Check whether the secret is mounted as a volume or injected as an environment variable.
- 4. Validate pod spec references the correct secret name and key.
- 5. Review service account, RBAC, external secret operator, and secret sync status if applicable.
- 6. Check file mount path, file permissions, and application config path.
- 7. Update the secret reference, RBAC binding, mount path, or external secret sync.
- 8. Restart pods if secret values are injected as environment variables.

Common Commands

```
kubectl config current-context
kubectl get pods -n <namespace> -o wide
kubectl describe pod <pod-name> -n <namespace>
kubectl logs <pod-name> -n <namespace> --tail=100
kubectl get events -n <namespace> --sort-by=.lastTimestamp
kubectl get deploy,svc,ingress,pvc -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Secret not accessible	Wrong secret name or namespace	Correct reference
Key missing	Secret data incomplete	Recreate or update secret key
Permission denied	RBAC or file permission issue	Fix role binding or mount permissions
Env value unchanged	Pod not restarted after secret update	Restart pod or deployment

Best Practices

1. Do not print secret values in logs.
2. Use managed secret rotation workflows.
3. Validate secret key names in deployment reviews.
4. Restart pods safely after env-based secret updates.

References

[Kubernetes Docs](#) | [Internal Tickets CSV](#) | [Cluster Events](#) | [Monitoring Dashboards](#) | [Application Logs](#) | [Platform Runbook](#)